Single generators of polynomial rings under composition: a characteristic-two dichotomy

— draft —

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Abstract

In Boolean logic a single operation, NAND, suffices to express every Boolean function. We ask the analogous question for polynomial rings: is there a single binary polynomial p such that every polynomial in two variables can be written as an iterated composition of p (starting from the variables and constants)? Over $\mathbb{Z}$ the answer is no: for every $p \in \mathbb{Z}[x,y]$, at least one of $x+y$, xy is inexpressible. More generally, there is a single polynomial generating all of $R[x,y]$ under composition (with variables and constants) if and only if 2 is invertible in R: when it is, the single operation $p(x,y) = x^2 - y$ generates everything; when it is not, generation fails through a residue field of characteristic 2. The main work is the negative result over fields of characteristic 2, proved by an infinite descent whose decreasing quantity is, informally, the number of times a would-be counterexample factors through the squaring map. All results have been checked formally.

1 Introduction

1.1 A Sheffer stroke for polynomial rings

Every Boolean function can be built from a single connective: NAND (the *Sheffer stroke*) [14, 10]. One-element generating sets of this kind appear throughout algebra and logic, and it is natural to ask for an analogue one level up, with the two-element Boolean domain replaced by a ring of polynomials.

So fix the ring $\mathbb{Z}$ and consider polynomials in two variables. We are allowed to start from the raw materials

$$x, \qquad y, \qquad \text{and every constant } c \in \mathbb{Z},$$

and we are given *one* binary polynomial $p(x,y)$ as our only "connective": whenever f and g have been built, we may build $p(f,g)$. Write $\mathrm{Clo}(p)$ for the set of all polynomials obtainable this way (a precise definition is in Section 2; Section 1.3 explains how this relates to the standard notion of a *clone* in universal algebra; no background from that subject is needed). Say that p *generates* $\mathbb{Z}[x,y]$ if $\mathrm{Clo}(p) = \mathbb{Z}[x,y]$.

Could such a p exist? Nothing obvious rules it out; a single polynomial has plenty of room to encode both an "additive" and a "multiplicative" mode, depending on what is substituted into it. And since every polynomial is built from variables and constants by $+$ and $\times$, generation is equivalent to just two memberships (Lemma 2.2):

$$\mathrm{Clo}(p) = \mathbb{Z}[x, y] \iff x + y \in \mathrm{Clo}(p) \text{ and } xy \in \mathrm{Clo}(p).$$

Our first result is that no single polynomial achieves even this.

Theorem 1.1. *For every $p \in \mathbb{Z}[x, y]$, at least one of $x + y$, xy lies outside* $\mathrm{Clo}(p)$*. In particular, no binary polynomial generates $\mathbb{Z}[x, y]$.*

1.2 Results

Theorem 1.1 is the case $R = \mathbb{Z}$ of a dichotomy over arbitrary commutative rings, which is the main result of this paper:

Theorem 1.2 (the dichotomy)**.** *Let R be a commutative ring. There is a single polynomial generating all of $R[x, y]$ under composition (with variables and constants) if and only if 2 is a unit of R.*

Half of this is easy: over any commutative ring in which 2 is invertible, the operation

$$p(x, y) \;=\; x^2 - y$$

generates the full polynomial ring — a dozen lines of explicit algebra, presented in Section 3, valid over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}[\frac{1}{2}]$, and $\mathbb{F}_p$ for every odd prime p. (Over $\mathbb{F}_3$, for example, iterating $x^2 - y$ from the variables and constants produces *every* polynomial in $\mathbb{F}_3[x, y]$.)

Theorem 1.3. *Let R be a commutative ring with $\frac{1}{2} \in R$. Then $q(x, y) := x^2 - y$ generates the full polynomial ring:* $\mathrm{Clo}(x^2 - y) = R[x, y]$.

The same derivation, run over $\mathbb{Z}$, fails exactly at the steps that divide by 2; Theorem 1.1 says this failure is no accident of one derivation.

The negative half starts from the observation that generation survives reduction of coefficients: a ring homomorphism $R \to S$ carries any derivation of $x+y$ over R — by a *derivation* we mean a formula building $x+y$ from the variables and constants by repeated application of p — to one over S. Reducing mod 2, Theorem 1.1 therefore follows from the corresponding statement over the two-element field $\mathbb{F}_2 = \mathbb{Z}/2$:

Theorem 1.4. *For every $q \in \mathbb{F}_2[x, y]$,*

$$x + y \notin \mathrm{Clo}(q) \qquad \textit{or} \qquad xy \notin \mathrm{Clo}(q).$$

Two remarks on the statement. First, polynomials here are *formal* objects, not functions: over $\mathbb{F}_2$ infinitely many polynomials induce the same function on $\mathbb{F}_2^2$, and Theorem 1.4 concerns the polynomials themselves, so it does not follow from (and does not contradict) the classical theory of functionally complete operations on finite sets; Section 1.3 dwells on the distinction. Second — and this matters for the reduction — there is *no hypothesis on q whatsoever*. Reduction mod 2 can collapse degree ($2x^3 + xy \mapsto xy$), so to deduce Theorem 1.1 one needs the $\mathbb{F}_2$-statement for arbitrary q, including the low-degree and degenerate ones.

Passing to $\mathbb{F}_2$ is not merely a convenience: characteristic 2 is where the obstruction lies, and not only over the prime field. The descent behind Theorem 1.4 generalizes to every *perfect* field of characteristic 2 (one in which every element has a square root — all finite fields $\mathbb{F}_{2^k}$ and all algebraically closed fields, for instance), at the cost of a single coefficient twist explained in Section 6, and the disjunctive conclusion of Theorem 1.4 holds verbatim there; an arbitrary field of characteristic 2 then embeds into its perfection, and non-membership in the clone descends along coefficient embeddings. Hence:

Theorem 1.5. *For every field k of characteristic 2 and every $q \in k[x, y]$, at least one of $x + y$, xy lies outside $\mathrm{Clo}(q)$. In particular, over a field k of characteristic 2 no single binary polynomial generates $k[x, y]$.*

Finally, if 2 is not a unit of R, it lies in a maximal ideal; a generator would push forward along the quotient map to the residue field, which has characteristic 2. Section 6 assembles these reductions into a proof of Theorem 1.2.

1.3 What exactly is claimed

Because the objects here are formal polynomials rather than functions, some care is needed about what Theorems 1.1–1.5 do and do not assert; this subsection collects the necessary precision.

In universal algebra, a *clone* on a set A is a family of finitary operations $A^n \to A$ containing all coordinate projections and closed under composition; the classical theory — Post's lattice and its descendants — studies such families of *functions* [10, 16]. The elements of $\mathrm{Clo}(p)$ are instead formal polynomials, not maps. In clone-theoretic terms: the binary terms in the language of commutative rings with constant symbols form the binary part of an abstract clone (equivalently, of a Lawvere theory [6]), canonically identified with $R[x, y]$ under substitution, and our question asks when the single term p generates it. (Restricting to the binary part loses nothing: any term built from p, constants, and two variables is itself a bivariate polynomial, whatever arities its subterms pass through.) This abstract object is represented faithfully by operations on R when R is an infinite integral domain, and is finer than its image otherwise; it should not be confused with the clone of polynomial *operations* of the ring R in the sense of the universal-algebra literature [4, 16], nor with the full clone of all finitary operations on the set R. A reader who prefers to avoid clones altogether can take the low-tech view of Definition 2.1 at face value: $R[x, y]$ carries the single binary operation $(f, g) \mapsto p(f, g)$, and $\mathrm{Clo}(p)$ is the subalgebra of this magma generated by $\{x, y\} \cup R$.

Over an infinite integral domain a polynomial is determined by the function it induces, so over $\mathbb{Z}$ or $\mathbb{Q}$ our statements can equally be read as statements about operations (on the infinite set $\mathbb{Z}$ or $\mathbb{Q}$, where the classical finite-domain theory is silent). Over finite fields the formal question is strictly finer than the functional one:

Example 1.6. The distinction between formal polynomials and the functions they induce matters over finite fields. For example, $n(x, y) := 1 + xy \in \mathbb{F}_2[x, y]$ induces the binary NAND operation on $\mathbb{F}_2$, and every Boolean function is a composition of NAND; nevertheless $x + y \notin \mathrm{Clo}(n)$ by Theorem 1.4, since $Dn = 1 \neq 0$ for the mixed-derivative operator D of Section 4.

Two further cautions about nearby terminology. First, we admit all constants as generators; in the function-clone world this corresponds to *completeness with constants* in the classical functional sense. (The constant-free variant is also natural; we do not pursue it here.) Second, the phrase *polynomial completeness* has an established meaning in universal algebra — that every congruence-compatible function on an algebra is induced by a polynomial, as in the monograph [4] — unrelated to the generation property studied here.

1.4 The shape of the negative proof, in words

The substantial result is Theorem 1.4, and the rest of this introduction previews its proof at leisure; the new idea is the descent in step 3, and we will take it slowly again when it appears for real in Section 5.

Step 1: a derivative splits the problem. Characteristic 2 kills second derivatives in each single variable ($\partial_x^2 = \partial_y^2 = 0$) but not the mixed one, so the operator $D := \partial_x \partial_y$ carries unusual weight (all derivatives in this paper are formal partial derivatives of polynomials). We split according to Dq.

If $Dq = 0$ — that is, no monomial of q has both exponents odd — then a chain-rule identity shows the property "$Dg = 0$" is preserved by every substitution into q. The variables and constants have it; xy does not ($D(xy) = 1$). So $xy \notin \mathrm{Clo}(q)$, and this case is done.

If $Dq \neq 0$ we will show $x + y \notin \mathrm{Clo}(q)$; this case takes the rest of the argument.

Step 2: from clones to curves. To exclude $x+y$ we use an invariant of a softer kind: a rigidity property of substitution. We will show that, when $Dq \neq 0$, a substitution $q(\alpha, \beta)$ can only land on a nonconstant polynomial in $x + y$ if α and β are themselves polynomials in $x + y$. A short induction over the clone (Proposition 4.3) shows this rigidity implies $x + y \notin \mathrm{Clo}(q)$.

The rigidity property, in turn, is a statement about the *level sets* of q — the loci $\{q = c\}$, c a constant. Restricting everything to a generic line of the pencil $\{x + y = \text{const}\}$ of parallel lines converts it into the following concrete geometric assertion: there is no pair of one-variable polynomials $\alpha(t), \beta(t)$, not both constant, with

$$q\big(\alpha(t), \beta(t)\big) = c$$

identically in t, where c is a "generic" constant — precisely, a constant transcendental over $\mathbb{F}_2$ (satisfying no nonzero polynomial relation with $\mathbb{F}_2$-coefficients), living in a large algebraically closed field K of characteristic 2; the setting is fixed precisely in Section 4.3. Such a pair (α, β) would be a polynomial-parametrized curve lying inside a single generic level set of q; we call it a *witness* (Definition 4.5). The task is to show witnesses do not exist when $Dq \neq 0$.

Step 3: the descent. Here is the idea in one paragraph. Differentiate the witness equation $q(\alpha, \beta) = c$ with respect to t. In characteristic 2, derivatives interact with squares in extreme ways (squares have derivative zero; conversely, over a perfect field, anything with derivative zero *is* a square), and the derivative calculus along the witness forces a great deal of square structure. The endgame is a fork. Either some polynomial relation with $\mathbb{F}_2$-coefficients holds along the witness — and any such relation is fatal, because it pins the witness to algebraic data, contradicting the genericity of the level c — or no such relation holds, in which case both α and β turn out to be squares of polynomials of half the degree, $\alpha = \alpha_1^2$, $\beta = \beta_1^2$, and because the coefficients of q lie in $\mathbb{F}_2$ (where squaring fixes everything),

$$q(\alpha_1, \beta_1)^2 = q(\alpha_1^2, \beta_1^2) = c,$$

so (α_1, β_1) is again a witness, for the *same* q, at the level $\sqrt{c}$, with *half the degree*. Repeat. The degree cannot halve forever, so eventually the first branch occurs.

The induction, note, is not on q at all: q never changes. It is on the witness, and the decreasing quantity is the number of times the witness can still factor through the Frobenius (squaring) map — what one might call its *Frobenius depth*. This inversion — inducting on the counterexample's Frobenius depth rather than on any complexity measure of q — is the main idea of the proof; our attempts at induction on degree-like measures of q had all failed. Degree of q is a poor induction variable here because substitution composes degrees in uncontrollable ways (note e.g. that q may reproduce low-degree polynomials from high-degree inputs); Frobenius depth of the witness is immune to this.

Because the argument has an unusual density of characteristic-2 bookkeeping, we have verified all results formally; a short note on this appears in Section 8, and the text otherwise presents proofs as ordinary mathematics. Specifically, every result is checked down to the proof kernel in Lean, and the headline theorems are additionally verified by an adversarial judge (`leanprover/comparator`) against a standalone statement file of roughly ninety lines, with every proof replayed through two independent kernels — so the many edge cases, where an argument this dense is most fragile, are covered by the verification.

2 Definitions and conventions

Definition 2.1. Let R be a commutative ring, $R[x, y]$ the ring of formal polynomials in two variables, and $p \in R[x, y]$. The set $\mathrm{Clo}(p) \subseteq R[x, y]$ is the smallest set of polynomials such that

1. $x \in \mathrm{Clo}(p)$, $y \in \mathrm{Clo}(p)$, and $c \in \mathrm{Clo}(p)$ for every constant $c \in R$;
2. if $f, g \in \mathrm{Clo}(p)$ then $p(f, g) \in \mathrm{Clo}(p)$,

where $p(f, g)$ denotes substitution of f for x and g for y in p. We say p *generates* $R[x, y]$ if $\mathrm{Clo}(p) = R[x, y]$.

Thus $\mathrm{Clo}(p)$ is the set of polynomials expressible by a formula in the variables and constants whose only operation symbol is p.

Lemma 2.2. $\mathrm{Clo}(p) = R[x, y]$ *if and only if* $x + y \in \mathrm{Clo}(p)$ *and* $xy \in \mathrm{Clo}(p)$.

Proof. One direction is trivial. For the other, first note that substituting members of $\mathrm{Clo}(p)$ into a *member* of $\mathrm{Clo}(p)$ again lands in $\mathrm{Clo}(p)$ (induction on the formula). So if $x + y$ and xy are members, $\mathrm{Clo}(p)$ is closed under the derived binary operations $(f, g) \mapsto f + g$ and $(f, g) \mapsto fg$; since it contains the variables and constants, it contains every polynomial. $\square$

Lemma 2.3 (transfer)**.** *Let $\varphi : R \to S$ be a ring homomorphism and $\pi : R[x, y] \to S[x, y]$ the induced map on coefficients. Then $\pi\big(\mathrm{Clo}(p)\big) \subseteq \mathrm{Clo}(\pi p)$ for every $p \in R[x, y]$.*

Proof. π fixes the variables, sends constants to constants, and commutes with substitution; induct over the formula. $\square$

In particular, if both $x+y, xy \in \mathrm{Clo}(p)$ over $\mathbb{Z}$, then both lie in $\mathrm{Clo}(\pi p)$ over $\mathbb{F}_2$. So Theorem 1.4 (for *arbitrary* $q = \pi p$, degenerate cases included) implies Theorem 1.1, and likewise its analogue over any ring with a homomorphism to $\mathbb{F}_2$ ($\mathbb{Z}/2^k$, $\mathbb{F}_2$-algebras, ...). Sections 4 and 5 take place entirely over $\mathbb{F}_2$; general coefficients return in Section 6.

We will constantly use three elementary facts about the formal partial derivatives ∂_x, ∂_y on $\mathbb{F}_2[x,y]$, all special to characteristic 2:

1. *Squares are derivative-free*: $\partial(f^2) = 2f\,\partial f = 0$.
2. *Repeated differentiation in one variable dies*: $\partial_x^2 = \partial_y^2 = 0$ (a monomial surviving ∂_x has had its odd x-exponent made even), while the mixed operator $D := \partial_x\partial_y$ does not.
3. *Frobenius*: squaring is a ring homomorphism, $(f+g)^2 = f^2+g^2$; and a polynomial whose exponents are all even is a square (halve the exponents and use that every coefficient, being 0 or 1, is its own square root).

One further convention. Degrees of polynomials in one variable are taken with the convention that constants — including the zero polynomial — have degree 0; under this convention $\deg(f^2) = 2\deg f$ holds without exception, which keeps the bookkeeping of the descent in Section 5 honest.

3 The positive half: $x^2 - y$

We begin with the positive half of the dichotomy, which is self-contained and shows concretely what the negative half must rule out.

Proof of Theorem 1.3. By Lemma 2.2 it suffices to reach $x+y$ and xy. All steps below are instances of one identity:

$$q\big(\alpha,\, q(\alpha',\gamma)\big) = \alpha^2 - \alpha'^2 + \gamma. \tag{1}$$

Translations. With constants $\alpha = \frac{d+1}{2}$, $\alpha' = \frac{d-1}{2}$, identity (1) sends $\gamma \mapsto \gamma + d$, for any prescribed $d \in R$. (This is the only step that uses $\frac{1}{2}$.) In particular $x+d, y+d \in \mathrm{Clo}(q)$ for all d.

Slanted lines. $q(c,y) = c^2 - y$, then $q(x, c^2-y) = x^2 + y - c^2$, then

$$q\big(x+d,\ x^2+y-c^2\big) = 2dx - y + (d^2+c^2) \ \in\ \mathrm{Clo}(q),$$

a line of slope $2d$ for every d, with constant adjustable by translations.

Linear engine. For any $\alpha \in \mathrm{Clo}(q)$ and constant k, apply (1) with outer argument $\alpha + k$ and inner argument α: the increment is

$$\gamma \ \longmapsto\ (\alpha+k)^2 - \alpha^2 + \gamma \ =\ \gamma + 2k\,\alpha + k^2.$$

Increments along slanted lines of two distinct slopes span all linear polynomials, so starting from $\gamma = 0$: *every linear polynomial lies in* $\mathrm{Clo}(q)$, in particular $x+y$ and $x + \frac{y}{2}$.

Scaled squares. The same engine with $\alpha = y^2+\ell$ (in the clone via $q(y,-\ell)$) adds λy^2+const for any λ; from $\gamma = x^2 = q(x,0)$ we get $x^2 + \frac{1}{4}y^2 \in \mathrm{Clo}(q)$.

Finish.

$$q\Big(x+\tfrac{y}{2},\ x^2+\tfrac{1}{4}y^2\Big) = \Big(x+\tfrac{y}{2}\Big)^2 - x^2 - \tfrac{1}{4}y^2 = xy. \qquad \square$$

Remark 3.1. All constants used lie in $\mathbb{Z}[\frac{1}{2}]$, so the derivation runs verbatim over every ring in which 2 is invertible — in particular over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}[\frac{1}{2}]$, and $\mathbb{F}_p$ for odd p; over $\mathbb{F}_3$, the single operation $x^2 - y$ generates the full polynomial ring $\mathbb{F}_3[x, y]$. Over $\mathbb{Z}$ the translation step requires realizing $d = c^2 - e^2$ with $c, e \in \mathbb{Z}$, which is possible only for $d \not\equiv 2 \pmod 4$; the increments available are then too even to span, and the derivation breaks down — as, by Theorem 1.1, every derivation must.

Remark 3.2. Theorem 1.3 produces a generator of degree 2. We do not know the full picture for higher degrees; see Section 9.

4 The negative half over $\mathbb{F}_2$

We turn to the negative half. This section and the next prove Theorem 1.4. Fix $q \in \mathbb{F}_2[x, y]$.

4.1 Parity decomposition and the derivative dichotomy

Group the monomials $x^i y^j$ of q into four classes by the parities of (i, j). Each class is an even-exponent polynomial times one of $1, x, y, xy$, and even-exponent polynomials are squares (fact 3 of Section 2). Hence q can be written, uniquely, as

$$q \;=\; A_0^2 \;+\; xA_1^2 \;+\; yA_2^2 \;+\; xy\,A_3^2, \qquad A_i \in \mathbb{F}_2[x, y]. \tag{2}$$

Differentiating (2) and discarding the squares (fact 1), only the prefixes x, y, xy contribute:

$$\partial_x q = A_1^2 + yA_3^2, \qquad \partial_y q = A_2^2 + xA_3^2, \qquad Dq = A_3^2. \tag{3}$$

So Dq is always a perfect square, and $Dq = 0$ exactly when q has no monomial with both exponents odd — equivalently, when $q \in \mathbb{F}_2[x^2, y] + \mathbb{F}_2[x, y^2]$, i.e. when q partially factors through the squaring map. The polynomial A_3 measures the failure of this degeneration, and the whole proof is organized around it.

The degenerate case $Dq = 0$ is settled by a conserved quantity:

Lemma 4.1 (chain rule for D)**.** *In characteristic* 2*, for all* $q, \alpha, \beta \in \mathbb{F}_2[x, y]$*:*

$$D\big(q(\alpha, \beta)\big) = (Dq)(\alpha, \beta)\,\big(\partial_x\alpha\,\partial_y\beta + \partial_y\alpha\,\partial_x\beta\big) + (\partial_x q)(\alpha, \beta)\,D\alpha + (\partial_y q)(\alpha, \beta)\,D\beta.$$

Proof sketch. Expand $\partial_x\partial_y\big(q(\alpha, \beta)\big)$ by the ordinary chain and product rules. The terms that would involve $\partial_x^2 q$ or $\partial_y^2 q$ vanish by fact 2, and the survivors assemble as displayed. (In characteristic 0 the same computation produces additional second-derivative terms; their absence here is the point.) □

Proposition 4.2. *If* $Dq = 0$ *then* $xy \notin \mathrm{Clo}(q)$*.*

Proof. The set $\{g : Dg = 0\}$ contains the variables and constants, and by Lemma 4.1 (with the first term dead) it is closed under substitution into q. Hence it contains all of $\mathrm{Clo}(q)$. But $D(xy) = 1$. □

From now on suppose $Dq \neq 0$, i.e. $A_3 \neq 0$; we must show $x + y \notin \mathrm{Clo}(q)$.

4.2 The rigidity property (†) and the master reduction

Write $\mathbb{F}_2[x+y]$ for the subring of polynomials in $x + y$ (the "diagonal" polynomials). The proof of Theorem 1.4 in the case $Dq \neq 0$ runs through the following property of q:

> (†) for all $\alpha, \beta \in \mathbb{F}_2[x, y]$: whenever $q(\alpha, \beta)$ is a *nonconstant* element of $\mathbb{F}_2[x+y]$, both $\alpha, \beta \in \mathbb{F}_2[x+y]$.

In words: the only way a substitution into q can land on a nonconstant diagonal polynomial is for both inputs to be diagonal already. The property is engineered to feed the following induction.

Proposition 4.3 (master reduction)**.** *If q satisfies* (†) *then $x + y \notin \mathrm{Clo}(q)$.*

Proof. We claim every $g \in \mathrm{Clo}(q)$ satisfies

$$P(g): \qquad g \notin \mathbb{F}_2[x+y] \quad \text{or} \quad g \text{ is constant.}$$

The variables satisfy P: the substitution $x \leftrightarrow y$ fixes every element of $\mathbb{F}_2[x+y]$ but moves x, so $x \notin \mathbb{F}_2[x+y]$, likewise y. Constants satisfy P. Inductive step: let $g = q(\alpha, \beta)$ with $P(\alpha)$, $P(\beta)$, and suppose $P(g)$ fails: $g \in \mathbb{F}_2[x+y]$, nonconstant. Property (†) forces $\alpha, \beta \in \mathbb{F}_2[x+y]$; then $P(\alpha), P(\beta)$ force α, β constant; but then g is constant, a contradiction. So P holds on the clone, and $x + y$ — which is in $\mathbb{F}_2[x+y]$ and nonconstant — is not in it. □

So Theorem 1.4 reduces to:

Theorem 4.4. *If $Dq \neq 0$ then q satisfies* (†)*.*

4.3 From polynomials to curves

Property (†) becomes a geometric statement after restriction to a generic member of the pencil of lines $\{x + y = \text{const}\}$. Fix once and for all an algebraically closed field $K \supseteq \mathbb{F}_2$ of characteristic 2 containing an element σ transcendental over $\mathbb{F}_2$; concretely, $K = \overline{\mathbb{F}_2(\sigma)}$, an algebraic closure of the field of rational functions in one indeterminate. We say $c \in K$ is *transcendental* if it satisfies no nonzero polynomial relation over $\mathbb{F}_2$ — the algebraic stand-in for "c is a generic value".

Definition 4.5. A *witness* is a pair $\alpha, \beta \in K[t]$, not both constant, such that $q(\alpha, \beta) = c$ is a constant, with c transcendental.

Geometrically: a polynomial-parametrized curve $t \mapsto (\alpha(t), \beta(t))$ in the plane over K, lying entirely inside the single level set $\{q = c\}$ of a generic level c.

Proposition 4.6 (bridge)**.** *If no witness exists, then q satisfies* (†)*.*

Proof. Let $\alpha, \beta \in \mathbb{F}_2[x, y]$ with $q(\alpha, \beta) = f(x + y)$, $f \in \mathbb{F}_2[u]$ nonconstant. Substitute the parametrized generic line $(x, y) = (t, t + \sigma)$, under which $x + y \mapsto \sigma$. The left side becomes $q\big(\alpha(t, t{+}\sigma),\ \beta(t, t{+}\sigma)\big)$ and the right side the *constant* $f(\sigma)$, which is transcendental (a nonzero $\mathbb{F}_2$-relation on $f(\sigma)$ composes with f to a nonzero relation on σ). Since no witness exists, both $\alpha(t, t{+}\sigma)$ and $\beta(t, t{+}\sigma)$ are constant in t. It remains to see that a polynomial $g \in \mathbb{F}_2[x, y]$ that is constant along the generic line lies in $\mathbb{F}_2[x+y]$: expand $g(x,\, x + u) =$

$\sum_i g_i(u)\, x^i$ with $g_i \in \mathbb{F}_2[u]$; constancy in t of $g(t, t+\sigma)$ says $g_i(\sigma) = 0$ for all $i \geq 1$, hence $g_i = 0$ (σ transcendental), so $g(x, x+u) = g_0(u)$, i.e. $g = g_0(x+y)$ after substituting $u = y-x$ (which in characteristic 2 is $y + x$). □

Theorem 4.4 has thus become a concrete statement we can attack with one-variable calculus:

Theorem 4.7 (no witnesses)**.** *If $Dq \neq 0$, no witness exists.*

5 The Frobenius descent

This section proves Theorem 4.7. We first develop the mechanism in the simplest nontrivial situation, then do the general case.

Throughout, (α, β) is a putative witness: $\alpha, \beta \in K[t]$ not both constant, $q(\alpha, \beta) = c$, c transcendental. A prime $'$ denotes d/dt. Degrees are degrees in t, under the convention of Section 2 (constants, including the zero polynomial, have degree 0). Recall the two calculus facts, now over $K[t]$: squares have zero derivative; and conversely, since K is algebraically closed (hence *perfect*: every element has a square root), a polynomial in $K[t]$ with zero derivative is a square — it is a polynomial in t^2, and its coefficients have square roots.

5.1 Warm-up: the case $A_3 = 1$

Suppose first the parity decomposition (2) has $A_3 = 1$:

$$q = A_0^2 + xA_1^2 + yA_2^2 + xy, \qquad \partial_x q = A_1^2 + y, \quad \partial_y q = A_2^2 + x.$$

(The smallest example is $q = xy$ itself; the case already contains all polynomials whose unique both-exponents-odd monomial is xy.)

Differentiate the witness equation $q(\alpha, \beta) = c$ in t. The chain rule gives

$$A\,\alpha' + B\,\beta' = 0, \qquad A := (\partial_x q)(\alpha, \beta), \quad B := (\partial_y q)(\alpha, \beta). \tag{4}$$

Next differentiate A and B themselves. Using $\partial_x(\partial_x q) = 0$ and $\partial_y(\partial_x q) = Dq = 1$ (and symmetrically):

$$A' = \beta', \qquad B' = \alpha'. \tag{5}$$

Combining (4) and (5):

$$(AB)' = A'B + AB' = \beta' B + A\alpha' = 0,$$

so AB is a *square* in $K[t]$. This is the engine: along a witness, the product of the two partial derivatives is forced to be a perfect square.

Now the fork. (For this warm-up we take for granted that neither A nor B vanishes identically; this degeneracy, like several others in the general case, is dispatched by the kill lemma below — Lemma 5.1 — and we do not belabor it here.) Suppose first A and B have a common root $\theta \in K$. Then the point $P := (\alpha(\theta), \beta(\theta))$ satisfies $\partial_x q(P) = \partial_y q(P) = 0$: it is a *critical point* of q. In the warm-up case the critical equations read $A_1^2(P) = \beta(\theta)$ and $A_2^2(P) = \alpha(\theta)$ — two polynomial equations over $\mathbb{F}_2$ pinning P down; one shows (this is easy here, and is Lemmas 5.2 and 5.3 in general) that P must have coordinates *algebraic* over $\mathbb{F}_2$. But then $c = q(P)$ is algebraic — contradicting transcendence of the level. A generic level set passes through no critical point.

Otherwise A and B are coprime in $K[t]$. A product of two coprime polynomials is a square only if each factor is (up to unit, and units of $K[t]$ are constants of the algebraically closed K, hence squares). So A and B are squares; squares have zero derivative; and (5) turns this into

$$\alpha' = \beta' = 0.$$

By perfectness, $\alpha = \alpha_1^2$ and $\beta = \beta_1^2$ for some $\alpha_1, \beta_1 \in K[t]$ of *half* the degree. Here the $\mathbb{F}_2$-coefficients enter: squaring fixes 0 and 1, so squaring a polynomial commutes with substituting into q,

$$q(\alpha_1, \beta_1)^2 \;=\; q(\alpha_1^2,\, \beta_1^2) \;=\; q(\alpha, \beta) \;=\; c.$$

Writing $r := \sqrt{c} \in K$ (transcendental, since r algebraic would make $c = r^2$ algebraic), we conclude $q(\alpha_1, \beta_1) = r$: *the pair (α_1, β_1) is again a witness, for the same q, at half the total degree.*

Iterating, the total degree $\deg \alpha + \deg \beta$ halves at every round in the second branch, and cannot do so forever; the first branch must eventually occur, giving the contradiction. This proves Theorem 4.7 in the warm-up case. The induction variable, to repeat the point from the introduction, is a property of the *witness* — how many more times it can factor through squaring, its Frobenius depth — while q stands still.

5.2 What changes in general

For general q with $Dq = A_3^2 \neq 0$, two new things happen.

First, the engine identities (5) acquire the factor $\Delta := (Dq)(\alpha, \beta) = A_3(\alpha, \beta)^2$:

$$A' = \Delta\, \beta', \qquad B' = \Delta\, \alpha',$$

which is harmless *provided* $\Delta \neq 0$ along the witness — a new way the argument can degenerate.

Second and more seriously, the fork's first branch needs critical points of q to be algebraic, and this can simply be false: the critical equations $\partial_x q = \partial_y q = 0$ may have *curves* of solutions. Indeed if $w := \gcd(A_1, A_2, A_3)$ is nonconstant then by (3) the whole curve $\{w = 0\}$ consists of critical points, and a witness might pass through such points. The remedy is to peel w off before running the engine. Write $A_i = wB_i$, so that the B_i have no common factor, and

$$q = A_0^2 + w^2\, h, \qquad h := xB_1^2 + yB_2^2 + xyB_3^2, \tag{6}$$

with $B_3 \neq 0$. We will run the warm-up argument *on h instead of q*: the function $H := h(\alpha, \beta)$ is no longer constant along the witness, but its derivative still vanishes, which is what makes the peel legal. From (6) and $q(\alpha, \beta) = c = r^2$:

$$W^2 H = S^2, \qquad W := w(\alpha, \beta), \quad S := r + A_0(\alpha, \beta), \tag{7}$$

using $(u + v)^2 = u^2 + v^2$. If $W \neq 0$, differentiating (7) (both W^2 and S^2 are squares) gives $W^2 H' = 0$, so $H' = 0$. The engine only ever used the vanishing of the level's derivative, so it runs on h unchanged.

What the peel buys is the algebraicity of critical points: for h, unlike q, the critical locus contains no curve. That is the keystone lemma below. Finally, the various nondegeneracy conditions accumulated along the way ($W \neq 0$, $\Delta \neq 0$, $A \neq 0$, $B \neq 0$) are all of the same shape — some fixed nonzero polynomial with $\mathbb{F}_2$-coefficients vanishing identically along the witness — and a single lemma disposes of them all: no such relation can hold, because it would tie the generic level to algebraic data.

We now state the three lemmas and assemble the proof.

5.3 Three lemmas

Lemma 5.1 (kill lemma)**.** *Let (α, β) be a witness with level c, and $f \in \mathbb{F}_2[x, y]$, $f \neq 0$. Then $f(\alpha, \beta) \neq 0$ in $K[t]$.*

Proof sketch. Suppose $f(\alpha, \beta) = 0$; after swapping variables if needed, assume α nonconstant. Adjoin a fresh variable z standing for the level, and consider f and $g := q + z$ as polynomials in y over the ring $\mathbb{F}_2[x, z]$. Since g is monic and linear in z, it is irreducible and cannot divide f (which is z-free), so f and g are coprime; a Bézout identity over the fraction field, with denominators cleared, produces

$$f\,U + (q + z)\,V = E(x, z) \neq 0 \qquad \text{in } \mathbb{F}_2[x, z][y].$$

Now substitute $y \mapsto \beta$, $x \mapsto \alpha$, $z \mapsto c$: the first summand dies by hypothesis, the second because $q(\alpha, \beta) + c = c + c = 0$ in characteristic 2. So $E(\alpha, c) = 0$. Expanding $E = \sum_i \rho_i(z)\, x^i$ and using that $1, \alpha, \alpha^2, \ldots$ have strictly increasing degrees in t (here nonconstancy of α enters), each coefficient $\rho_i(c)$ must vanish; some $\rho_i \neq 0$ then exhibits c as algebraic over $\mathbb{F}_2$ — contradiction. □

Lemma 5.2 (algebraic points)**.** *If $u, v \in \mathbb{F}_2[x, y]$ have no common nonunit factor and $(a, b) \in K^2$ satisfies $u(a, b) = v(a, b) = 0$, then a, b are algebraic over $\mathbb{F}_2$.*

Proof sketch. Viewing u, v in $\mathbb{F}_2(x)[y]$: a common divisor there would descend (Gauss's lemma; see e.g. [5, Ch. IV]) to a common factor in $\mathbb{F}_2[x, y]$, so they are coprime, and a cleared Bézout identity gives $su + tv = r(x) \neq 0$; evaluate at (a, b) to get $r(a) = 0$. Symmetrically for b. □

Lemma 5.3 (keystone: no curve of critical points after the peel)**.** *Let $B_1, B_2, B_3 \in \mathbb{F}_2[x, y]$ have no common nonunit factor, with $B_3 \neq 0$, and set $h_x := B_1^2 + yB_3^2$, $h_y := B_2^2 + xB_3^2$ (the partials of h). Then h_x and h_y have no common nonunit factor.*

Proof. Both are nonzero: applying ∂_y to a relation $h_x = 0$ would give $B_3^2 = 0$. Suppose a prime p divides both.

If $p \mid B_3$: subtracting the B_3^2-terms, p divides B_1^2 and B_2^2, hence (primality) all three B_i — excluded.

If $p \nmid B_3$: write $h_x = pu$ and $h_y = pv$. Differentiate $h_x = B_1^2 + yB_3^2$, remembering squares are derivative-free:

$$0 = \partial_x h_x = p_x u + p\, u_x \quad \Longrightarrow \quad p \mid p_x u;$$

$$B_3^2 = \partial_y h_x = p_y u + p\, u_y \quad \Longrightarrow \quad p \nmid u$$

(else $p \mid B_3^2$, so $p \mid B_3$);

$$B_3^2 = \partial_x h_y = p_x v + p\, v_x \quad \Longrightarrow \quad p \nmid p_x$$

(same reason). So the prime p divides the product $p_x u$ but neither factor — contradiction. $\square$

Remark 5.4. Lemma 5.3 has geometric content: a curve of common zeros of h_x, h_y would have a function field in which both x and y become squares (solve the critical equations for y and x), making that function field *perfect*; but the function field of a curve in characteristic 2 is never perfect. The displayed proof is this argument in infinitesimal form — "$p \nmid p_x$" is the derivation that survives — and entirely avoids the geometry.

5.4 Proof of Theorem 4.7

Among all witnesses, take one minimizing $n := \deg \alpha + \deg \beta$ (≥ 1, as not both are constant). Decompose q as in (6); fix $r = \sqrt{c}$, transcendental.

By the kill lemma (Lemma 5.1) applied to each of the nonzero polynomials w, h_x, h_y, B_3 — nonzero by construction and by the argument opening Lemma 5.3 — none of

$$W := w(\alpha, \beta), \quad A := h_x(\alpha, \beta), \quad B := h_y(\alpha, \beta), \quad \Delta := B_3(\alpha, \beta)^2$$

vanishes in $K[t]$.

Peel. As in (7), $W^2 H = S^2$ with $H = h(\alpha, \beta)$, $S = r + A_0(\alpha, \beta)$, and differentiating, $H' = 0$.

Engine. The chain rule for h, plus $\partial_x h_x = 0$, $\partial_y h_x = \partial_x h_y = Dh = B_3^2$, $\partial_y h_y = 0$, gives

$$A\alpha' + B\beta' = H' = 0, \qquad A' = \Delta\beta', \qquad B' = \Delta\alpha',$$

whence $(AB)' = \Delta(A\alpha' + B\beta') = 0$ and AB is a square.

Fork, branch 1: A and B share a root θ. Then $P := (\alpha(\theta), \beta(\theta))$ is a common zero of h_x, h_y, which by Lemmas 5.2 and 5.3 has algebraic coordinates. Evaluate the peel identity (7) at θ:

$$S(\theta)^2 = w(P)^2\, h(P)$$

is algebraic, hence so is $S(\theta)$, hence so is $r = S(\theta) + A_0(P)$. This contradicts transcendence of r.

Fork, branch 2: A, B coprime. Coprime factors of a square are squares, so $A' = B' = 0$; since $\Delta \neq 0$, the engine identities force $\alpha' = \beta' = 0$, so $\alpha = \alpha_1^2$, $\beta = \beta_1^2$, and as in the warm-up $q(\alpha_1, \beta_1) = r$. Then (α_1, β_1) is a witness with $\deg \alpha_1 + \deg \beta_1 = n/2 < n$, contradicting minimality. (If $n = 1$ this branch cannot occur at all, as halving is impossible; so the descent is well-founded.) $\square$

Assembling: Theorem 4.7 $\Rightarrow$ (bridge, Proposition 4.6) property (†) $\Rightarrow$ (master reduction, Proposition 4.3) $x + y \notin \mathrm{Clo}(q)$ when $Dq \neq 0$; together with Proposition 4.2 for $Dq = 0$, this proves Theorem 1.4, and with Lemma 2.3, Theorem 1.1. $\square$

6 From $\mathbb{F}_2$ to the dichotomy

Theorem 1.4 is the kernel of the negative half; the rest of Theorem 1.2 follows from it by three short reductions, which we now record.

Step 1: perfect fields, via a coefficient twist.

Proposition 6.1. *Let k be a perfect field of characteristic 2. For every $q \in k[x,y]$, at least one of $x+y$, xy lies outside $\mathrm{Clo}(q)$.*

Proof. The descent generalizes to any perfect coefficient field k of characteristic 2 with one modification. The coefficients of q are no longer fixed by squaring, so in the halving branch the identity $q(\alpha_1^2, \beta_1^2) = \tilde{q}(\alpha_1, \beta_1)^2$ holds not for q itself but for the *coefficient-twisted* polynomial $\tilde{q} = q^{(1/2)}$ (apply the inverse Frobenius to each coefficient — this is where perfectness enters). The halving branch therefore produces a half-degree witness for $q^{(1/2)}$ rather than for q; since the induction is on the witness degree with q universally quantified inside, the twist is absorbed at no cost, and no periodicity argument is needed. The remaining ingredients — the parity decomposition, the master reduction, and the bridge — are unaffected, and the disjunctive conclusion follows as over $\mathbb{F}_2$. $\square$

Step 2: arbitrary fields of characteristic 2, via the perfection.

Proof of Theorem 1.5. An arbitrary field k of characteristic 2 embeds into its perfection $k^{1/2^\infty}$, which is perfect of characteristic 2. The coefficient embedding is a ring homomorphism, so by the transfer lemma (Lemma 2.3) a derivation of $x+y$ (or xy) over k maps to one over the perfection; if both $x+y$ and xy lay in $\mathrm{Clo}(q)$ over k, both would lie in the clone over the perfection, contradicting Proposition 6.1. $\square$

Step 3: rings in which 2 is not a unit, via residue fields.

Proof of Theorem 1.2. If 2 is a unit of R, the operation $x^2 - y$ generates $R[x,y]$ by Theorem 1.3. If 2 is not a unit, it lies in a maximal ideal $\mathfrak{m}$, and a generator over R would push forward along $R \to R/\mathfrak{m}$ (Lemma 2.3 again) to a generator over the residue field, which has characteristic 2 — contradicting Theorem 1.5. $\square$

7 Relation to classical functional completeness

For operations-as-functions on a finite set, one-element functional completeness is classical territory. Post's classification of the Boolean clones locates the Sheffer stroke [14, 10]; Webb exhibited a single functionally complete operation on every finite domain [17]; and there are effective functional completeness criteria, from Słupecki through Rousseau to Rosenberg's classification of the maximal function clones [15, 13, 12].

Example 1.6 (Section 1.3) showed that over finite fields the formal statements of this paper and the classical functional ones can diverge. In the positive direction, however, the generation theorem does project onto a classical statement, which we record for universal algebraists. Recall that an algebra is *functionally complete* if every finitary operation on its underlying set is a polynomial operation (= a term operation after all constants are adjoined).

Corollary 7.1. *Let F be a finite field of odd order. Then the algebra $\langle F;\ x^2 - y\rangle$ with all constants adjoined is functionally complete: every function $F^n \to F$, for every n, is a composite of $x^2 - y$, constants, and variables.*

Proof. By Theorem 1.3 (proved in Section 3; 2 is invertible in F), $x+y$ and xy are composites of $x^2 - y$ and constants, hence so is every polynomial in any number of variables; and by Lagrange interpolation every function $F^n \to F$ is induced by a polynomial. □

Functional completeness criteria for finite algebras are classical territory, from Foster's primal algebras and the Foster–Pixley theory of semi-primal algebras [2, 3] to Werner's discriminator varieties [18]; Corollary 7.1 places $\langle F;\ x^2 - y\rangle$ (with constants) inside that landscape, whereas Theorem 1.4, by Example 1.6, lives strictly outside it.

The composition structure of polynomial rings has its own literature — Menger's tri-operational algebras, Adler's composition rings, and near-ring theory for univariate composition [8, 1, 9], as well as Ritt's theory of decompositions of a single univariate polynomial [11]. These study the composition structure of the full polynomial ring rather than generation from a single binary operation, and we have not found the present question addressed there. We would be grateful for pointers to antecedents; the question is natural enough that we hesitate to claim novelty for anything but the answers.

8 A note on formal verification

The proofs above, especially the descent, involve a large amount of routine characteristic-2 manipulation, and we wanted an independent check. All results stated here — Theorems 1.1, 1.2, 1.3, 1.4 and the supporting lemmas — have been verified in the Lean 4 proof assistant against the Mathlib library, with no unproved assumptions beyond Lean's standard foundational axioms. The formal statements involve only the fifteen-line definition of Clo reproduced in Appendix A. In addition, the headline theorems are certified by an adversarial judge (`leanprover/comparator`): their statements live in a standalone challenge file of roughly ninety lines that does not import the development, and the judge replays all proofs through two independent proof kernels against those exact statements, so trusting the results requires reading only the challenge file and the verdict. Two of the proofs presented above are actually simplifications found during the verification process: the derivative proof of the keystone (Lemma 5.3), which replaced a longer argument via perfect function fields, and the Bézout-certificate form of the kill lemma. We will not say more about the formalization here; the development — including a blueprint mapping each informal statement of this paper to its formal declaration — is available from the author and will be made public with the paper.

9 Open questions

Degrees and characterization

Question 9.1. Which degrees admit single generators over $\mathbb{Q}$? Degree 2 does (Theorem 1.3). For degree 3 a natural candidate is $x^3 - y$: translations by arbitrary rationals follow from the classical fact that every rational number is a sum of three rational cubes (with explicit formulas), and a cubic analogue of the linear engine appears to reach xy; we have not finished the verification. The answer is characteristic-sensitive: over $\mathbb{F}_3$ the map $u \mapsto u^3$ is additive (Frobenius), and one checks that $\mathrm{Clo}(x^3 - y)$ consists only of polynomials of the form *additive plus constant* — $\mathbb{F}_3$-linear combinations of the monomials x^{3^i}, y^{3^j}, plus a constant (the constants enter because they are generators) — a class closed under substitution into $x^3 - y$ and excluding xy; so $x^3 - y$ is certainly not a generator there.

Question 9.2. Away from characteristic 2, *which* operations are single generators? Generation is not simply "nonlinear and depending on both variables": over $\mathbb{Q}$,

- $xy + 1$ is not a generator, although it is nonlinear, uses both variables, and $\mathrm{Clo}(xy + 1)$ contains xy. The obstruction is a characteristic-zero instance of the rigidity property (†): if $\alpha\beta + 1 = F(x+y)$ is nonconstant then $\alpha\beta = h(x+y)$ splits over $\overline{\mathbb{Q}}$ into linear factors $(x+y - r_i)$, so by unique factorization every divisor — in particular α and β — lies in $\mathbb{Q}[x+y]$; the master reduction (whose proof is characteristic-free) then excludes $x + y$.
- $x^2 + y$ is not a generator, although $x^2 - y$ is: every element of $\mathrm{Clo}(x^2 + y)$ has the form $a + \sum_i s_i^2$ with $a \in \{x, y\} \cup \mathbb{Q}$ (the shape is preserved by $(f, g) \mapsto f^2 + g$), and neither $x + y$ nor xy is of this form, since sums of squares of real polynomials take no negative values. A *positivity* obstruction from the real place, invisible over finite fields.

Together with the affine obstruction (linear p) and the monomial obstruction ($\mathrm{Clo}(xy)$ consists of monomials), this suggests asking whether generation is governed by a finite explicit list of "conserved classes" — proper subsets of $R[x, y]$ containing the generators and closed under substitution into p. We do not know; we also cannot presently decide simple boundary cases such as $x^2 - 2y$, where the engine increment $\gamma \mapsto 4\gamma + (\alpha^2 - 2\alpha'^2)$ rescales instead of translating and a 2-adic conserved class is conceivable. A positive structure theorem here would likely also bear on Question 9.3.

Decidability

Question 9.3. Is membership in $\mathrm{Clo}(p)$ decidable? Is generation?

Half of the problem is easy: membership is semidecidable, since one can enumerate all formulas in p and expand each. It is the complementary half that we want to dwell on, because the difficulty is structural and shaped our whole approach. The obvious idea — search all formulas up to some size or degree bound, and conclude non-membership if the target never appears — proves nothing, no matter the bound, because substitution is not monotone in any evident complexity measure. The culprit is *cancellation*. Already for $q = x^2 + y$ over $\mathbb{F}_2$ one has, for *all* f, g,

$$q\big(f,\ q(f, g)\big) \;=\; f^2 + (f^2 + g) \;=\; g\,:$$

the intermediate polynomial f, of arbitrarily large degree, is annihilated without trace. So every element of the clone admits derivations passing through intermediates of unbounded degree, and a small target may, for all a bounded search can tell, first become reachable only after a long excursion through enormous polynomials that cancel at the last step. The degree of what you are building says nothing about the degree of what you must build it from.

This is why every non-membership argument in this paper avoids search entirely and instead exhibits a *cancellation-immune invariant*: a property that is checked on the final polynomial alone and is preserved by every substitution into q, whatever the intermediate terms are — membership in $\ker D$ in Proposition 4.2, the invariant P in Proposition 4.3. A decision procedure would seem to require either a computable bound on derivation size, which cancellation undermines, or a supply of such invariants rich enough to certify every non-membership; we possess neither, and we suspect the question is hard. For comparison: in the cousin problem of membership in the *ideal* generated by given polynomials, the analogous degree question is settled — doubly exponential degree bounds make ideal membership decidable, indeed EXPSPACE-complete by Mayr–Meyer [7] — and it is exactly such a bound that cancellation withholds here. As a concrete instance of how little we know: for q with $Dq = 0$, Theorem 1.4 settles $xy \notin \mathrm{Clo}(q)$, but we know no algorithm deciding, for such q, whether $x + y \in \mathrm{Clo}(q)$.

Universal algebra

Question 9.4. The universal-algebra side of the picture is wide open beyond Corollary 7.1, which adjoins all constants. Without constants: is $\langle F;\, x^2 - y\rangle$ *primal* for every finite field F of odd order — is every finitary operation on F a term operation of $x^2 - y$ alone? (Rousseau's criterion [13] reduces this to a concrete finite check for any one F; we have not pursued it.) More broadly, we have attempted no Mal'cev-condition analysis of the algebras $\langle R;\, p\rangle$: which congruence conditions (permutability, distributivity, ...) the varieties they generate satisfy, and whether these interact with p generating $R[x, y]$, are natural questions for universal algebraists that our methods do not touch.

Rational functions

Question 9.5. The same questions can be posed for the field of rational functions: preliminary calculations suggest $1/x - y$ is a single generator of $\mathbb{Q}(x, y)$ in a suitable sense. Substitution of rational functions is a partial operation (denominators may vanish identically), so even stating this precisely requires a partial-clone formalism.

Draft note. All references below have been checked against the published record, with one exception: the issue and page numbers of the Menger item could not be confirmed against the original (the series and year are confirmed); it is marked in the source. Further pointers, especially to antecedents of the main question, remain welcome.

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A The formal statement layer

For the record, the full definitional trust base of the formal verification (R an arbitrary commutative ring; `X 0`, `X 1` the two variables of a bivariate polynomial ring; `bind1 ![a,b] p` denotes the substitution $p(a, b)$; the formal development uses `complete` in declaration names for the generation property):

```
inductive Clo (p : MvPolynomial (Fin 2) R) :
    MvPolynomial (Fin 2) R -> Prop
  | atomX : Clo p (X 0)
  | atomY : Clo p (X 1)
  | const (c : R) : Clo p (C c)
  | sub {a b : MvPolynomial (Fin 2) R} :
      Clo p a -> Clo p b -> Clo p (bind1 ![a, b] p)

theorem F2_master (q : MvPolynomial (Fin 2) (ZMod 2)) :
    ~ Clo q (X 0 + X 1) \/ ~ Clo q (X 0 * X 1)

theorem int_master (p : MvPolynomial (Fin 2) Int) :
    ~ Clo p (X 0 + X 1) \/ ~ Clo p (X 0 * X 1)

theorem qOp_complete {R} [CommRing R] [Invertible (2 : R)] :
    forall g, Clo (X 0 ^ 2 - X 1 : MvPolynomial (Fin 2) R) g

theorem complete_iff_two_isUnit (R) [CommRing R] :
    (exists p : MvPolynomial (Fin 2) R, forall g, Clo p g)
      <-> IsUnit (2 : R)
```